EXPERIMENT ON CISCO PACKET TRACER

Exp no: 3 Date:

Aim:

To study the installation of Cisco Packet Tracer, understand its user interface layout and analyse the operational differences between network hubs and switches by building simple star topologies.

Software Requirements:

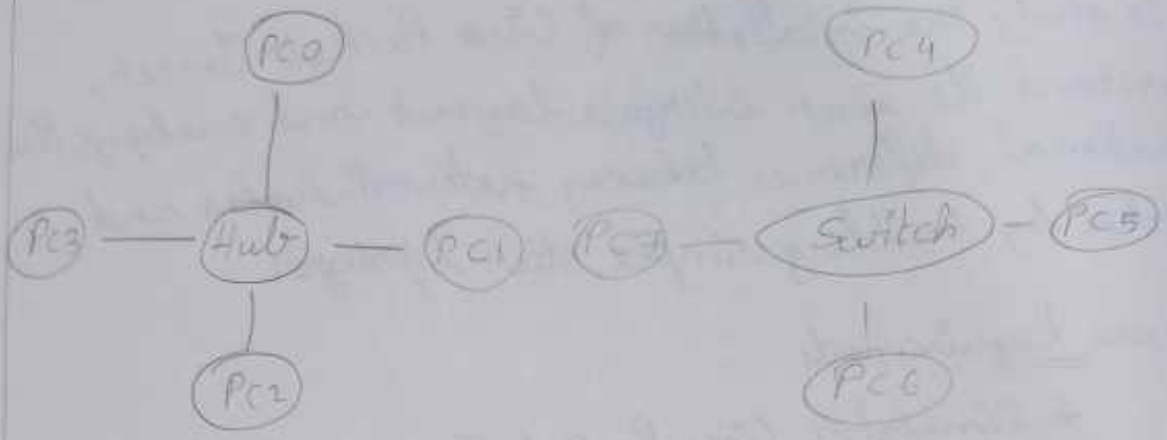
* Simulator: Cisco Packet Tracer
* OS support: Windows / Linux
* Installation file: Packettracer_setup 6.0.1 exe

User Interface Breakdown:

UI component	Interface Region	Description / Function
1	Menu bar	Contains top level menus for file operations and system preferences
2	Main Toolbar	Quick access icons for open, save, print, zoom.
3	Logical/Physical workspace tab	Toggles between the logical network topology view and the physical geographical view
4	Workspace	The primary canvas area where devices are placed, connected and simulated
5	Common Tools bar	Utilities for interacting with the workspace
6	Real time Tabs	Switches mode between live network behaviour and step-by-step packet inspection

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Simulation Topologies & Setup procedure:



1. Device Placements:

- Network A: Place 1 Generic Hub and 4 Generic PCs (PC0, PC1, PC2, PC3)
- Network B: Place 1 Generic Switch and 4 Generic PCs (PC4, PC5, PC6, PC7)

2. Cabling:

- Connect all PCs to their respective central device using copper - straight - through cables. Verify green of link LEDs on connected ports.

3. IP Address Configuration:

- Navigate to Desktop → IP configuration on each PC.
- Assign static PC IP address within the same subnet.

4. PDU traffic test:

- Select the Add Simple PDU tool
- Set a source PC and a target destination PC
- Switch to Simulation Mode to observe packet transmission across the medium.

F.

Device Behavior Analysis & Conclusion:

- Hub:

* Observation: When a PDU is sent from one PC to another via the Hub, the Hub broadcasts the incoming frame out of all active ports except the receiving port.

- Switch:

* Observation: When a PDU is sent through the Switch, after learning MAC addresses, it forwards the frame only to the specific port connected to the intended destination PC.